

Each question is worth 5 points for a total of 115 points. The extra credit can get you an additional 5 points making the highest possible score 120 out of 115. Write answers CLEARLY. If I cannot determine your answer, the problem will receive a score of 0.

Problem 1

Find the exact value $\frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ}$.

Problem 2

Find the exact value $\sin[\sin^{-1} \frac{3}{5} - \cos^{-1}(-\frac{4}{5})]$.

Problem 3

Find the exact value $\tan \frac{7\pi}{12}$.

Problem 4

Establish the identity and SHOW ALL YOUR WORK.

$$\tan(3\theta) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

Problem 5

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin \theta + 1 = 2$.

Problem 6

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin^2 \theta = 1 + \sin \theta$.

Problem 7

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\cos(2\theta) = \cos \theta$.

Problem 8

To measure the height of Lincoln's caricature on Mt. Rushmore, two sightings 400 feet from the base of the mountain are taken. If the angle of elevation to the bottom of Lincoln's face is 45° and the angle of elevation to the top is 60° , what is the height of Lincoln's face?

Problem 9

Solve the triangle $a = 9$, $b = 18$, $\alpha = 30^\circ$.

Problem 10

Find the area of the triangle $a = 6$, $b = 18$, $\gamma = \frac{\pi}{3}$.

Problem 11

Find the area of the triangle $a = 3$, $b = 3$, $c = 3$.

Problem 12

The displacement d (in meters) of an object at time t (in seconds) is given by the equation $d = 5 \cos(\pi t)$. Find the amplitude, period, and frequency.

Problem 13

Convert each angle in degrees to radians. Express your answer as a multiple of π .

- a) 60°
- b) 270°
- c) 45°
- b) 120°

Problem 14

Let s denote the length of the arc of a circle of radius r subtended by the central angle θ . Find the missing quantity.

- a) $r = 16$ meters, $\theta = 2$, $s = ?$
- b) $r = 3$ inches, $\theta = ?$, $s = \pi$ inches
- c) $r = ?$, $\theta = \frac{\pi}{6}$, $s = 2\pi$ feet

Problem 15

Find the exact value.

- a) $\sin \frac{13\pi}{6}$
- b) $\cos \frac{7\pi}{4}$
- c) $\tan \frac{3\pi}{9}$

Problem 16

The point $(5, 12)$ is on the terminal side of an angle θ . Find the exact value of the six trigonometric functions.

Problem 17

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \cos^2 \theta + \cos \theta - 1 = 0$.

Problem 18

Let $\sin \theta = \frac{3}{4}$ where θ is in the first quadrant. Find the exact value of the other five trigonometric functions.

Problem 19

Find the exact value of $\cot(85^\circ)\tan(85^\circ)$.

Problem 20

Determine the amplitude and period of $y = -4\sin(\pi x)$.

Problem 21

Graph $y = 2\csc(x - \frac{\pi}{2})$.

Problem 22

Find the exact value of $\csc(\cos^{-1}\frac{3}{5})$.

Problem 23

Establish the identity and SHOW ALL YOUR WORK.

$$\frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2.$$

Problem 24

Show that $\sec(\tan^{-1} v) = \sqrt{1 + v^2}$.

Extra Credit

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\sqrt{3} \sin \theta + \cos \theta = 1$.